

In [1]: `import pandas as pd`

In [2]: `ssim_df = pd.read_csv('ssim.csv')`
`ssim_df`

Out[2]:

	category	content	SSIM parameters set 1	SSIM parameters set 2	SSIM parameters set 3	SSIM parameters set 4	SSIM parameters set 5
0	landscape	beaulieu	0.209986	0.054180	0.022443	0.153979	0.169131
1	landscape	beaulieu	0.214029	0.080033	0.025743	0.157171	0.334289
2	landscape	beaulieu	0.303134	0.172214	0.028252	0.266713	0.335573
3	still-life	skull	0.101914	0.051823	0.015587	0.091950	0.429772
4	still-life	skull	0.305680	0.189936	0.020415	0.297561	0.344851
5	still-life	skull	0.316816	0.195207	0.023217	0.297387	0.343552
6	portrait	girl	0.415137	0.260297	0.034836	0.411860	0.462338
7	portrait	girl	0.295497	0.136942	0.031002	0.328436	0.339268
8	portrait	girl	0.230094	0.102864	0.029521	0.216930	0.219434
9	portrait	man	0.235351	0.102899	0.027693	0.179743	0.241078
10	portrait	man	0.355389	0.156908	0.032182	0.315319	0.411497
11	portrait	man	0.465034	0.275813	0.033390	0.392104	0.449830
12	landscape	sea	0.477864	0.337286	0.036081	0.557083	0.541419
13	landscape	sea	0.188139	0.116533	0.028481	0.213139	0.504453
14	landscape	sea	0.091457	0.088491	0.024113	0.093289	0.094173
15	landscape	mountain	0.123257	0.062465	0.020705	0.118160	0.126175
16	landscape	mountain	0.386705	0.228977	0.028673	0.396105	0.453346
17	landscape	mountain	0.181916	0.094070	0.023798	0.155231	0.411111
18	still-life	still	0.414461	0.317972	0.036049	0.425128	0.448248
19	still-life	still	0.145544	0.080609	0.023099	0.139929	0.498059
20	still-life	still	0.358233	0.175674	0.030938	0.397688	0.427335
21	portrait	repin	0.264787	0.096421	0.025996	0.243571	0.281245
22	portrait	repin	0.402834	0.109913	0.029996	0.401873	0.445047
23	portrait	repin	0.478213	0.224162	0.030181	0.448358	0.508120
24	still-life	fish	0.216354	0.073500	0.023937	0.193486	0.230553
25	still-life	fish	0.211661	0.145739	0.028447	0.170802	0.220414
26	still-life	fish	0.159535	0.055543	0.020580	0.133579	0.247276

In [3]: `ssim_df['ssim1_normalized'] = ssim_df['SSIM parameters set 1'] / ssim_df['SSIM parameters set 1'].mean() - 1`
`ssim_df`

Out[3]:

	category	content	SSIM parameters set 1	SSIM parameters set 2	SSIM parameters set 3	SSIM parameters set 4	SSIM parameters set 5	ssim1_normalized
0	landscape	beaulieu	0.209986	0.054180	0.022443	0.153979	0.169131	-0.248960
1	landscape	beaulieu	0.214029	0.080033	0.025743	0.157171	0.334289	-0.234500
2	landscape	beaulieu	0.303134	0.172214	0.028252	0.266713	0.335573	0.084196
3	still-life	skull	0.101914	0.051823	0.015587	0.091950	0.429772	-0.635492
4	still-life	skull	0.305680	0.189936	0.020415	0.297561	0.344851	0.093303
5	still-life	skull	0.316816	0.195207	0.023217	0.297387	0.343552	0.133132
6	portrait	girl	0.415137	0.260297	0.034836	0.411860	0.462338	0.484789
7	portrait	girl	0.295497	0.136942	0.031002	0.328436	0.339268	0.056881
8	portrait	girl	0.230094	0.102864	0.029521	0.216930	0.219434	-0.177042
9	portrait	man	0.235351	0.102899	0.027693	0.179743	0.241078	-0.158237
10	portrait	man	0.355389	0.156908	0.032182	0.315319	0.411497	0.271093
11	portrait	man	0.465034	0.275813	0.033390	0.392104	0.449830	0.663250
12	landscape	sea	0.477864	0.337286	0.036081	0.557083	0.541419	0.709140
13	landscape	sea	0.188139	0.116533	0.028481	0.213139	0.504453	-0.327099
14	landscape	sea	0.091457	0.088491	0.024113	0.093289	0.094173	-0.672894
15	landscape	mountain	0.123257	0.062465	0.020705	0.118160	0.126175	-0.559155
16	landscape	mountain	0.386705	0.228977	0.028673	0.396105	0.453346	0.383097
17	landscape	mountain	0.181916	0.094070	0.023798	0.155231	0.411111	-0.349354
18	still-life	still	0.414461	0.317972	0.036049	0.425128	0.448248	0.482371
19	still-life	still	0.145544	0.080609	0.023099	0.139929	0.498059	-0.479442
20	still-life	still	0.358233	0.175674	0.030938	0.397688	0.427335	0.281264
21	portrait	repin	0.264787	0.096421	0.025996	0.243571	0.281245	-0.052958
22	portrait	repin	0.402834	0.109913	0.029996	0.401873	0.445047	0.440785
23	portrait	repin	0.478213	0.224162	0.030181	0.448358	0.508120	0.710389
24	still-life	fish	0.216354	0.073500	0.023937	0.193486	0.230553	-0.226185
25	still-life	fish	0.211661	0.145739	0.028447	0.170802	0.220414	-0.242968
26	still-life	fish	0.159535	0.055543	0.020580	0.133579	0.247276	-0.429405

In [4]: `ssim_df['ssim2_normalized'] = ssim_df['SSIM parameters set 2'] / ssim_df['SSIM parameters set 2'].mean() - 1`
`ssim_df['ssim3_normalized'] = ssim_df['SSIM parameters set 3'] / ssim_df['SSIM parameters set 3'].mean() - 1`
`ssim_df['ssim4_normalized'] = ssim_df['SSIM parameters set 4'] / ssim_df['SSIM parameters set 4'].mean() - 1`
`ssim_df['ssim5_normalized'] = ssim_df['SSIM parameters set 5'] / ssim_df['SSIM parameters set 5'].mean() - 1`

In [5]: `ssim_df.head(5)`

Out[5]:

	category	content	SSIM parameters set 1	SSIM parameters set 2	SSIM parameters set 3	SSIM parameters set 4	SSIM parameters set 5	ssim1_normalized	ssim2_normalized	ssim3_normalized	ssim4_normalized	ssim5_normalized
0	landscape	beaulieu	0.209986	0.054180	0.022443	0.153979	0.169131	-0.248960	-0.633042	-0.175944	-0.422305	-0.520199
1	landscape	beaulieu	0.214029	0.080033	0.025743	0.157171	0.334289	-0.234500	-0.457941	-0.054799	-0.410328	-0.051670
2	landscape	beaulieu	0.303134	0.172214	0.028252	0.266713	0.335573	0.084196	0.166388	0.037346	0.000649	-0.048027
3	still-life	skull	0.101914	0.051823	0.015587	0.091950	0.429772	-0.635492	-0.649009	-0.427709	-0.655024	0.219200
4	still-life	skull	0.305680	0.189936	0.020415	0.297561	0.344851	0.093303	0.286421	-0.250405	0.116385	-0.021708

In [6]: `norm_cols = ["ssim1_normalized", "ssim2_normalized", "ssim3_normalized", "ssim4_normalized", "ssim5_normalized"]`
`df_flat = ssim_df.melt(id_vars="category", value_vars=norm_cols, var_name="experiment", value_name="normalized_value")`

In [7]: `df_flat`

Out[7]:

	category	experiment	normalized_value
0	landscape	ssim1_normalized	-0.248960
1	landscape	ssim1_normalized	-0.234500
2	landscape	ssim1_normalized	0.084196
3	still-life	ssim1_normalized	-0.635492
4	still-life	ssim1_normalized	0.093303
...	...	...	...
130	portrait	ssim5_normalized	0.262534
131	portrait	ssim5_normalized	0.441463
132	still-life	ssim5_normalized	-0.345955
133	still-life	ssim5_normalized	-0.374719
134	still-life	ssim5_normalized	-0.298515

135 rows x 3 columns

In [8]: `df_rq1 = (df_flat.groupby("category")["normalized_value"].agg(["mean", "std", "median", "count"]))`
`df_rq1`

Out[8]:

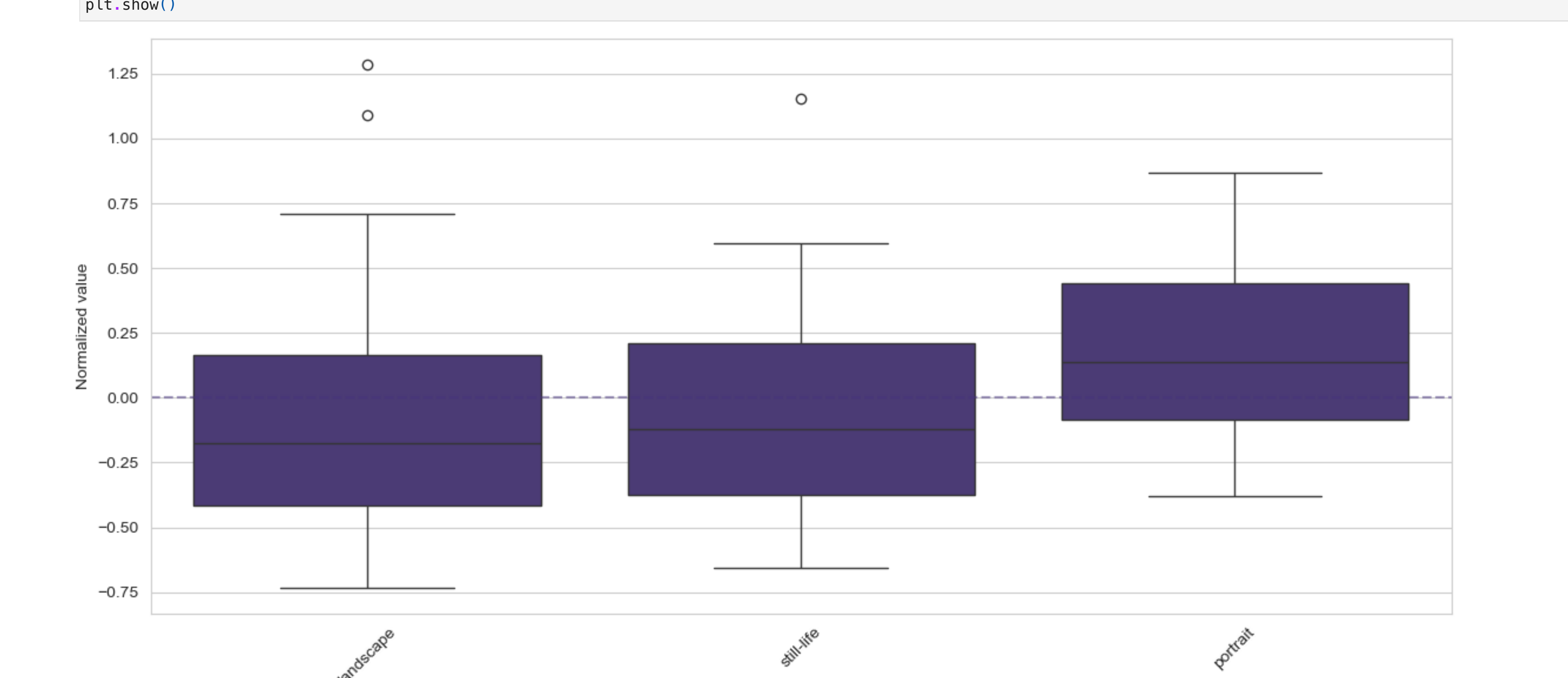
	mean	std	median	count
category				
landscape	-0.083616	0.463978	-0.175944	45
portrait	0.151298	0.333804	0.138285	45
still-life	-0.067682	0.392844	-0.121102	45

In [9]: `df_flat.groupby('category').describe()`

Out[9]:

	count	mean	std	min	25%	50%	75%	max
category								
landscape	45.0	-0.083616	0.463978	-0.732846	-0.417608	-0.175944	0.166262	1.284404
portrait	45.0	0.151298	0.333804	-0.377497	-0.086174	0.138285	0.440785	0.868058
still-life	45.0	-0.067682	0.392844	-0.655024	-0.374719	-0.121102	0.212286	1.153592

In [10]: `import seaborn as sns`
`import matplotlib.pyplot as plt`
`sns.set_palette("viridis")`
`plt.figure(figsize=(12, 6))`
`sns.boxplot(data=df_flat, x="category", y="normalized_value")`
`plt.axhline(0, linestyle="---", alpha=0.5)`
`plt.xticks(rotation=45)`
`plt.ylabel("Normalized value")`
`plt.xlabel("Category")`
`plt.tight_layout()`
`plt.savefig("category_results.png", dpi=300, bbox_inches="tight")`
`plt.show()`



In [11]: `sns.pointplot(data=df_flat, x="normalized_value", y="category", hue="experiment", errorbar=None, dodge=True)`
`plt.savefig("category_exp_results.png", dpi=300, bbox_inches="tight")`
`plt.axhline(0, linestyle="---", alpha=0.5)`

Out[11]: `<matplotlib.lines.Line2D at 0x16c40cc20>`

